

MC312 – Modeling and Simulation

Course Title: Modelling and Simulation

Credit Structure(L-T-P-Cr): (3-0-2-4)

Course Code: MC312

Prerequisites (if any)/desired skill set: Introductory Calculus, Probability and Statistics, Linear Algebra, **and basic programming is required.**

Level: Advanced Undergraduate

Course Placement: Modeling and Simulation is a core course offered to third year students of B.Tech Mathematics and Computing program.

Course format: It is **3 hours' lecture** and **2 hours lab** every week

Course content: This course introduces students to the fundamentals of creating mathematical models of physical systems and implementation on computers to analyze the system to gain insight and make predictions. The different mathematical approaches to modeling covered in the course can be characterized into differential and difference equation-based models, probability-based models, including stochastic differential equations, cellular automata, event-based approaches, and matrix-based models. The course is **interdisciplinary** and looks at systems from a modeling perspective from physics, biology, social science, engineering, etc. Examples from different disciplines follow each topic. The students are expected to test and create models of these systems in the lab, gaining intuitive understanding and reporting on it following the complete modeling approach. It may be the foundation for domain-based specialization electives such as Computational Finance, Computational, and Systems Biology and Computational Physics, Computational Social Science, etc.

Text Book:

Angela B. Shiflet & George W. Shiflet. *Introduction to Computational Science: Modeling and Simulation for the sciences*, Princeton University Press, 2006.

Reference Book:

- 1) Sheldon M. Ross, *Simulation*, Academic Press, 2006.
- 2) Hiroki Sayama, *Introduction to the Modeling and Analysis of Complex Systems*, Open SUNY Textbooks, Milne Library, 2015.
- 3) Taylor & Karlin, *An Introduction to Stochastic Modeling*, Academic Press, 1998.
- 4) S. Strogatz, *Nonlinear Dynamic and Chaos*, Oxford University Press, 2001.

Assessment method: Two in-semester examinations (40%), final examination (30%) and lab viva (30%).

Course Outcomes:

After completing the course, the students would be able to create a relevant model for many problems from science and engineering by extracting the necessary information regarding the situation. They would also be able to define the different modeling terms by analyzing the system or the available data. They would be able to implement the model on the computer and check for its validity and the correctness of the assumptions present in the model. They will analyze the outcomes (mainly through visualizations) and make predictions. They would also be able to understand the limitations of their model and nuances in the computer modeling of systems.

After successful completion of the course the student will have the ability to

- Apply the knowledge of mathematics and computer science to model complex problems in science (**P1,P2,P3,P12**).
- Carry out model based investigations to derive insight into the problem at hand (**P4**).
- Communicate effectively through vivas, course presentation and report writing (**P10**)

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X						X		X

Lecture Schedule *

Sl. No.	Description		No. of Lectures
1	Overview		2
	1.1	Introduction to Complex Systems	
	1.2	The Modeling Process.	
	1.3	Model Classification	
	1.4	Steps in the modeling Process.	
2	System Dynamics Model (Compartment Models)		15-18
	2.1	Unconstrained growth models.	
	2.2	Constrained Growth.	
	2.3	Applications: Radioactive decay, Drug dosage, population dynamics, growth of technologies, SI models for epidemic spread etc.	
	2.4	System of Equations:	

	2.5	Models from Epidemiology: epidemic spread models (SIR, SEIR, SEIZ etc.), predator-prey models, Modeling spread of infectious disease such as SARS, Malaria	
	2.6	Models from Social Science: Rumor spread, Models for Innovation Diffusion etc	
	2.7	Selective methods for analyzing one and two dimensional nonlinear differential equations	
3	Stochastic models (Monte Carlo simulations)		12-16
	3.1	Review of basic probability theory.	
	3.2	Generating random numbers from different distributions on computer.	
	3.3	Brownian motion and random walk in 1D, discrete and continuous time stochastic differential equation with applications.	
	3.4	Symmetric, asymmetric, Markovian and non-markovian random walk	
	3.5	Markov Chains, Continuous Time Markov Chains, Birth and death processes, limiting time behavior.	
4	Agent based Simulation (ABM)		8-12
	4.1	Cellular Automata simulation of random walk.	
	4.2	Ising Model for ferromagnetic Systems with applications to social science, metropolis algorithm.	
	4.3	Applications: Modeling opinion formation on 2D lattice, segregation models, Rumor spread models etc.	

* The number of topics covered will depend on the overall enthusiasm and commitment of the class